Chinese remaindering lowers the T-count of binary-field multipliers by up to two thirds

Claude Opus 5.5 Extra Effort[1]

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We report ancilla-free Clifford+T circuits for multiplication in GF(2^n) that use fewer T gates than the best published ones for twelve field sizes, $n = 6$, 7, 8, 10–16, 32 and 64. Examples are 69 against 77 T gates for $n = 6$, 97 against 101 for $n = 7$, 181 against 215 for $n = 11$, 571 against 1177 for $n = 32$, and 1255 against 3729 for $n = 64$. The circuits are built from a composition principle. If the product in GF(2^n) can be computed by Chinese remaindering in a product of small algebras A_i, then the T-count is at most $\sum_i \mathcal{T}(A_i)$, because every block is a linear relabelling of a small multiplier. The blocks can be subfields, local rings $\mathbb{F}_2[y]/(q^e)$ or the points of an algebraic curve. With optimized blocks, for example 15 T gates for $\mathbb{F}_2[y]/(y^2)$ (optimal by exhaustive search) and 35 for GF(4)$[\epsilon]/(\epsilon^2)$, the principle gives certified circuits for prime n, where subfield interpolation does not apply. A genus-2 curve with 33 rational points over GF(16) gives $n = 64$. FastTODD-type reduction and simulated annealing over the coset of decompositions remove further gates. For $n = 6$ and 7, annealing with reheating and slow cooling takes the composed starts from 85 to 69 and from 111 to 97 T gates. We also prove that 17 T gates are optimal for GF(4), and we report that the published 23 T gates for GF(8) could not be reproduced. Every circuit was certified by an exact verifier and by independent implementations, and by simulation of all inputs for each record with $n \leq 12$. The results were found by a multi-agent language-model system working against a frozen verifier.

I. INTRODUCTION

In fault-tolerant quantum computing with magic-state distillation, the T gate is the expensive gate, and its count is a standard cost metric [1, 2]. Minimizing the T-count of a circuit is hard in general. For circuits whose non-Clifford part is a phase polynomial it is equivalent to decoding a punctured Reed–Muller code, or to finding a minimal symmetric (Waring) decomposition of a third-order tensor over $\mathbb{F}_2$ [3, 4]. Heuristics for this problem include TODD [4] and its descendants FastTODD [5], reinforcement learning (AlphaTensor-Quantum) [6], tensor-decomposition methods [7], and LLM-guided evolutionary search (VarTODD+GigaEvo) [8]. Together they define the state of the art on standard benchmark circuits [9].

Multiplication in GF(2^n) is one of these benchmarks and a building block of quantum attacks on binary elliptic curves [10–12]. Its best published T-counts (Table I) come from three sources. Vandaele reports 23 for $n = 3$, from FastTODD applied to a Karatsuba-type circuit [5, 13]; we could not reproduce this value for GF(8) (Sec. VI). AlphaTensor-Quantum gives the best values for $n = 4$, 6 and 8 [6], which Fisher *et al.* match. VarTODD+GigaEvo gives the best for $n = 5$, 7, 9–16, 32 and 64 [8]; its value 101 for $n = 7$ starts from a decomposition of Ref. [7].

These methods treat each GF(2^n) as an unstructured tensor. We use the algebra instead. Classical algebraic complexity computes a product of polynomials by reducing it modulo several coprime polynomials, multiplying the residues and recombining them (Toom–Cook, Karatsuba, Winograd) [14–16]. Chudnovsky and Chudnovsky extended the idea to the points of algebraic curves [18]. We show that the T-count composes in the same way (Theorem 1). Every residue product is a linear relabelling of a multiplier in a small algebra, so the T-count of a composed circuit is the sum of the T-counts of its blocks. Optimized blocks then give good large circuits without any search on the large tensor, and FastTODD-type reduction or annealing improves them further.

An earlier version of this work used only subfield interpolation and covered six composite n. Here, residues modulo powers of irreducible polynomials (local rings) extend the method to prime n, an algebraic curve extends it to $n = 64$, and annealing improves the small fields $n = 6$ and 7. Section II fixes notation. Section III proves the composition principle, and Section IV gives its three instances. Section V describes the reduction and annealing. Section VI presents the records and the small-field status. Section VII describes the verification, and Section VIII how the results were found.

II. PRELIMINARIES

Let $P \in \mathbb{F}_2[x]$ be irreducible of degree n. Elements $a, b, c \in \mathrm{GF}(2^n) = \mathbb{F}_2[x]/(P)$ are coordinate vectors in $\mathbb{F}_2^n$. The in-place multiplier $|a, b, c\rangle \mapsto |a, b, c \oplus ab\rangle$ equals $H_c\, D\, H_c$, where H_c applies Hadamard gates to the c register and D is the diagonal gate

$$D = \sum_x (-1)^{f_n(x)} |x\rangle\langle x|, \qquad f_n(a, b, c) = c \cdot (ab), \quad (1)$$

where $x = (a, b, c)$ and $\cdot$ is the dot product over $\mathbb{F}_2$. The function f_n is a cubic form in the $N = 3n$ variables of x, whose monomials are $a_i b_j c_k$ for each x^k that appears in $x^{i+j} \bmod P$.

TABLE I. T-counts of ancilla-free GF(2^n) multipliers. "Previous" is the best published value: Ref. [8] (Table I) for $n \neq 6, 8$, and Refs. [6, 8] for $n = 6$ and 8. "Start" is the composed circuit of Sec. IV: K is the base field and the blocks are listed with their T-counts (Table II). Here ∞^e, t^e are rational places of multiplicity e, $p = y^2 + y + 1$, "deg k" is an irreducible place of degree k (a block GF(2^{mk})), and ω generates GF(4). "Search" is the step from the start to the final count: FastTODD-type reduction (FT) or annealing (SA) (Sec. V).

n	Previous	This work	K	Start: blocks	Raw	Search
6	77	**69**	GF(4)	5 points: 5×17	85	FT 79, SA 77, 71, 69
7	101	**97**	$\mathbb{F}_2$	$\infty^3, x^3, (x+1)^2$, deg 2, deg 3: 25+25+15+17+29	111	SA 100, 99, 97
8	123	**113**	GF(16)	3 points (Karatsuba): 3×39	117	FT
10	157	**151**	GF(4)	$\infty^2, 0^2, 1^2, \omega^2, \bar{\omega}$: $4 \times 35 + 17$	157	FT
11	215	**181**	$\mathbb{F}_2$	$\infty^3, x^2, (x+1)^2, p^2$, 2×deg 3, deg 4: 25+15+15+35+58+39	187	FT
12	243	**191**	GF(16)	5 points: 5×39	195	FT
13	293	**219**	$\mathbb{F}_2$	$\infty^3, x^2, (x+1)^2, p^2$, 2×deg 3, 2×deg 4: 25+15+15+35+58+78	226	FT
14	319	**229**	GF(4)	∞, $0^2, 1^2, \omega^2, \bar{\omega}^2$, 2×deg 2: 17+4 × 35+2 × 39	235	FT
15	363	**253**	GF(8)	9 points: 9×29	261	FT
16	383	**267**	GF(16)	7 points: 7×39	273	FT
32	1177	**571**	GF(16)	15 points: 15×39	585	FT
64	3729	**1255**	GF(16)	genus-2 curve, 33 points: 33×39	1287	FT

A T gate applied to the parity $u \cdot x$ (a phase gadget, conjugated by CNOT gates) contributes the phase $\omega^{\ell_u(x)}$, with $\omega = e^{i\pi/4}$ and $\ell_u(x) \in \{0, 1\}$ the parity as an integer. A set $\{u_1, \ldots, u_m\} \subset \mathbb{F}_2^N \setminus \{0\}$ *implements* D if D equals $\prod_r \omega^{\ell_{u_r}}$ times a diagonal Clifford gate. The T-count $\mathcal{T}(n)$ is the least such m. It does not depend on P or on the basis, because those changes are CNOT circuits.

Lemma 1 (Signature-tensor criterion [3, 4])**.** *$\{u_r\}$ implements D if and only if every variable and every pair of variables lies in the support of an even number of the u_r, and every triple lies in an odd number of them exactly when it is a monomial of f_n. Equivalently, $\sum_r u_r^{\otimes 3} \equiv S_n$ (mod 2), where S_n is the zero-diagonal symmetric tensor of f_n.*

Proof. Over $\mathbb{Z}_8$, $\ell_u(x) = \sum_{i \in u} x_i - 2\sum_{i<j \in u} x_i x_j + 4\sum_{i<j<k \in u} x_i x_j x_k$, and terms of degree ≥ 4 vanish because their coefficients are multiples of 8. Hence $4f_n - \sum_r \ell_{u_r}$ has linear coefficients $-\#\{r \ni i\}$, quadratic coefficients $2\#\{r \ni i, j\}$ and cubic coefficients $4(f_{ijk} - \#\{r \ni i, j, k\})$. A diagonal phase $\omega^{g(x)}$ is Clifford exactly when $g = 2h$ with h of degree ≤ 2 over $\mathbb{Z}_4$ and even quadratic coefficients (products of S and CZ gates). That is, the singles and pairs must be even and the triples must match f_n mod 2. □

The criterion is exact, so a set of vectors that passes it is a proof that the circuit is correct for all inputs; no sampling is involved. The same definitions apply to any finite commutative $\mathbb{F}_2$-algebra A with a chosen basis: $f_A(x, y, z) = z \cdot (xy)$ is a cubic form in $3 \dim A$ variables, and $\mathcal{T}(A)$ is its T-count. A change of basis, or of the nondegenerate pairing used for z, is an invertible linear substitution, so $\mathcal{T}(A)$ depends only on the isomorphism class of A. For example, $\mathcal{T}(\mathrm{GF}(2^k)) = \mathcal{T}(k)$.

III. COMPOSITION PRINCIPLE

Lemma 2 (Pull-back)**.** *Let $\{u_r\}$ implement the cubic form $g(y)$ on $\mathbb{F}_2^M$, and let $L : \mathbb{F}_2^N \to \mathbb{F}_2^M$ be linear. Then $\{L^{\mathsf{T}} u_r\}$, after deleting zero vectors and cancelling equal pairs, implements $g(Lx)$. Its size is at most the size of $\{u_r\}$.*

Proof. $\ell_u(Lx) = \ell_{L^{\mathsf{T}}u}(x)$, and Clifford gates on $y = Lx$ pull back to Clifford gates in x. Zero vectors and pairs u, u contribute Clifford phases ($\omega^{2\ell_u} = i^{\ell_u}$). □

Theorem 1. *Let $A_1, \ldots, A_s$ be finite commutative $\mathbb{F}_2$-algebras. Suppose there are $\mathbb{F}_2$-linear maps $\alpha_i, \beta_i : \mathrm{GF}(2^n) \to A_i$ and $\Gamma : A_1 \times \cdots \times A_s \to \mathrm{GF}(2^n)$ such that*

$$ab = \Gamma\big(\alpha_1(a)\beta_1(b), \ldots, \alpha_s(a)\beta_s(b)\big) \quad \text{for all } a, b. \tag{2}$$

Then $\mathcal{T}(n) \leq \sum_i \mathcal{T}(A_i)$.

Proof. Write $\Gamma(p_1, \ldots, p_s) = \sum_i \Gamma_i(p_i)$. For each i the map $p \mapsto c \cdot \Gamma_i(p)$ is $\mathbb{F}_2$-linear in $p \in A_i$, so it equals $\gamma_i(c) \cdot p$ for an $\mathbb{F}_2$-linear $\gamma_i : \mathbb{F}_2^n \to A_i$. Hence $f_n(a, b, c) = \sum_i f_{A_i}\big(\alpha_i(a), \beta_i(b), \gamma_i(c)\big)$. Lemma 2, applied to each term with $L_i = (\alpha_i, \beta_i, \gamma_i)$, gives a decomposition of size $\leq \sum_i \mathcal{T}(A_i)$. The union of the pulled-back sets implements the sum, because phases multiply and Clifford gates compose. □

The bilinear complexity of multiplication over $\mathbb{F}_2$ is linear in n [18, 19], and $\mathcal{T} \leq 7\times$(number of CCZ gates), so $\mathcal{T}(n) = O(n)$ was known. The point of Theorem 1 is the constant: it sums the blocks' *T-counts*, not $7\times$ their bilinear ranks. For GF(16) the bilinear rank is 9, so $7 \times 9 = 63$ against $\mathcal{T}(4) = 39$.

IV. THREE FAMILIES OF BLOCKS

Places of the projective line. Let $K = \mathrm{GF}(2^m)$ and $n = md$, and write $\mathrm{GF}(2^n) = K[y]/(g)$ with g irreducible of degree d over K. Elements are polynomials A, B of degree $< d$, and $C = AB$ has degree $\le 2d-2$. Choose pairwise coprime powers $q_i^{e_i}$ of monic irreducible $q_i \in K[y]$ and a multiplicity $e_\infty \ge 0$ at infinity, with

$$e_\infty + \sum_i e_i \deg q_i \;\ge\; 2d-1. \tag{3}$$

The residue $C \bmod q_i^{e_i}$ is the product of $A \bmod q_i^{e_i}$ and $B \bmod q_i^{e_i}$ in the ring $K[y]/(q_i^{e_i})$. The top e_∞ coefficients of C depend only on the top e_∞ coefficients of A and B, and they are their product in $K[y]/(y^{e_\infty})$ after reversing the coefficient order. These residues determine C: if they all vanish, C is divisible by $\prod_i q_i^{e_i}$ and has degree $\le 2d-2-e_\infty < \sum_i e_i \deg q_i$, so $C = 0$. Recovering C and reducing it modulo g is K-linear, which gives Eq. (2) with the blocks $K[y]/(q_i^{e_i})$ and $K[y]/(y^{e_\infty})$. Hence

$$\mathcal{T}(md) \le \mathcal{T}\big(K[y]/(y^{e_\infty})\big) + \sum_i \mathcal{T}\big(K[y]/(q_i^{e_i})\big). \tag{4}$$

A place of multiplicity one is a field, $K[y]/(q) \cong \mathrm{GF}(2^{m \deg q})$. A rational place of multiplicity two over K is the ring of dual numbers $K[\epsilon]/(\epsilon^2)$, and $\mathbb{F}_2[y]/(p^2)$ with $p = y^2 + y + 1$ is isomorphic to $\mathrm{GF}(4)[\epsilon]/(\epsilon^2)$.

With rational places of multiplicity one only, Eq. (4) is Toom–Cook interpolation: if $2^m \ge 2d-2$, the $2d-2$ finite points and infinity give $\mathcal{T}(md) \le (2d-1)\mathcal{T}(m)$, and $d = 2$ is Karatsuba's formula for every m [15–17]. Interpolation gives the starts for $n = 6$, 8, 12, 15, 16 and 32 in Table I. Prime n has no proper subfield, so $K = \mathbb{F}_2$, which has only three rational places (0, 1, ∞). For $K = \mathbb{F}_2$, the local rings make the difference. Each unit of degree costs 7.5 T gates in $\mathbb{F}_2[y]/(y^2)$ (15 for degree 2) but 8.5 in the field GF(4) (17 for degree 2). For the same reason, the dual numbers over GF(4) (35 T gates for 2 units) are cheaper than a degree-2 place over GF(4), the field GF(16) (39 for 2 units). A small dynamic program over the place multiset, with the block costs of Table II, chooses the starts for $n = 7$, 10, 11, 13 and 14.

An algebraic curve for $n = 64$. Over $K = \mathrm{GF}(16)$ with $d = 16$, Eq. (3) needs 31 units, but the projective line has only 17 rational places. The remaining 14 units must come from seven degree-2 places, each a GF(256) block of 113 T gates, so $17 \times 39 + 7 \times 113 = 1454$. Chudnovsky and Chudnovsky replace the line by a curve with more points [18]. The genus-2 curve $C: \; y^2 + y = x^5$ over GF(16) has 33 rational points. For each $x \in \mathrm{GF}(16)$, x^5 lies in GF(4), whose elements have trace zero in $\mathrm{GF}(16)/\mathbb{F}_2$, so each x gives two points; the point at infinity is the 33rd. This attains the Hasse–Weil bound $q + 1 + 2g\sqrt{q} = 33$. Let $D = 16P_\infty + P_a$ with P_a an affine rational point. By Riemann–Roch, $\dim L(D) = 16$ and $\dim L(2D) = 33 = 2d + g - 1$. For a suitable place Q of degree 16, whose residue field is $\mathrm{GF}(2^{64})$, evaluation at Q identifies $L(D)$ with $\mathrm{GF}(2^{64})$. At each rational point P, take the value of a function, or for $P \in \{P_\infty, P_a\}$ its leading Laurent coefficient. These 33 functionals are multiplicative and K-linear, so they give Eq. (2) with $A_i = K$ for every i, provided they are independent on $L(2D)$.

TABLE II. Block algebras and their T-counts $\mathcal{T}(A)$. "Exact" values come from exhaustive enumeration of the coset of decompositions (2^{22} elements). The others are certified upper bounds. $\mathcal{T}(\mathrm{GF}(8)) = 29$ is the best value we could certify; the published 23 is discussed in Sec. VI.

A	$\dim_{\mathbb{F}_2} A$	$\mathcal{T}(A)$	source
$\mathbb{F}_2$	1	7	CCZ, exact
$\mathbb{F}_2[y]/(y^2)$	2	15	this work, exact
$\mathbb{F}_2[y]/(y^3)$	3	25	this work
GF(4)	2	17	Refs. [5, 6]; exact (this work)
GF(8)	3	29	Ref. [6]
GF(16)	4	39	Refs. [6, 8]
$\mathrm{GF}(4)[\epsilon]/(\epsilon^2)$	4	35	this work
GF(64)	6	69	this work
$\mathrm{GF}(2^7)$	7	97	this work
GF(256)	8	113	this work

The shift by P_a is needed for independence. With $D = 17P_\infty$, the function $x^{16} - x$ vanishes at all 32 affine points and lies in $L(32P_\infty)$, so it is killed by all 33 functionals. With the shifted divisor, the kernel is $L(2D - E)$, where E is the sum of all rational points. Since $2D - E \sim 2P_a - P_\infty$ has degree 1, the kernel is nonzero only if $2P_a \sim P_\infty + R$ for some point R. Then $\dim L(2P_a) \ge 2$, so $2P_a$ would be the canonical class. That happens only when P_a is fixed by the hyperelliptic involution $y \mapsto y + 1$, and no affine point is. Shifted and asymmetric divisors of this kind are standard in Chudnovsky-type algorithms [20, 21]. The construction gives $\mathcal{T}(64) \le 33 \times 39 = 1287$, and reduction lowers it to 1255. Numerically, the 33×33 evaluation matrix is invertible for all 32 choices of P_a, and the product identity was checked on random inputs before the factors were emitted; the certified decomposition does not depend on these checks.

When curves help. A curve of genus g needs $2d + g - 1$ rational units instead of $2d - 1$. It beats the projective line only when the line runs out of points and the places that would replace them cost much more per unit than a rational one. Over GF(16), a degree-2 place costs 113 for 2 units, against $2 \times 39 = 78$. Over $\mathbb{F}_2$, GF(4) and GF(8) the local rings are cheap substitutes. A cost model with the blocks of Table II finds that no curve of positive genus beats the line for the other sizes in Table I. The same model predicts gains over GF(16) for $n = 44$, 48, 52, 56 and 60. These are estimates, not certified circuits.

V. REDUCTION AND ANNEALING

The decompositions of a fixed signature tensor form a coset of the code $\mathcal{C} = \{w \in \mathbb{F}_2^{2^N-1} : \sum_y w_y y^{\otimes 3} = 0\}$, the punctured Reed–Muller code $\mathrm{RM}(N-4, N)^*$ [3]. A composed circuit is one point of this coset, and its weight is the T-count. We lower the weight in two ways.

FastTODD-type reduction. Our C implementation of FastTODD [5] looks for a vector z and a set of factors such that adding z to each of them preserves the tensor and creates equal pairs, which cancel. It accepts neutral moves with a fixed probability and handles up to $N = 192$ variables. On the composed circuits it removes a few gates and then stops (Table I). For a circuit of b blocks it removes exactly $b-1$ gates at $n = 11$–16, 32 and 64, for example 32 for the 33 blocks at $n = 64$. The five-block circuits for $n = 6$ and 10 lose $b+1 = 6$, and the three-block circuit for $n = 8$ loses 4, but only in 3 of 10 seeds (the others end at 115). The pattern $\sum_k \mathcal{T}(A_k) - (b-1)$ mirrors the exact count $7m - (m-1) = 6m+1$ for m disjoint CCZ gates [23]. The smallest case, $\mathcal{T}(\mathbb{F}_2 \times \mathbb{F}_2) = 13 < 2 \times 7$, also follows from the exhaustive enumeration of Sec. VI. Blocks combined in one circuit can apparently share about one gate each. We state this only as an observation. The floors are robust: at $n = 11$, 12, 13, 16 and 64, all of 8 to 30 seeds and gauges (field isomorphisms and point choices) end at the same count.

Annealing over flats. For every affine 4-flat $F \subset \mathbb{F}_2^N$, $\sum_{v \in F} v^{\otimes 3} = 0$. The singles, pairs and triples restricted to F are polynomials of degree ≤ 3, and their sum over a 4-flat vanishes. So toggling the nonzero points of F in the decomposition is a move within the coset. We run Metropolis sampling on the count with proposals spanned by members of the current decomposition, so that many points of F are already present and the move is not too costly. At $n = 7$, a fixed temperature of 0.30 (in units of one gate) for 8×10^8 steps took the start 111 to 100. Restarting from each new best with a new seed at a slightly higher temperature (0.34, then 0.32) gave 99 and then 97. At $n = 6$, the start 85 went to 79 by FastTODD and to 77 at fixed temperature. Further fixed-temperature runs never left 77. A linear cooling schedule from 0.40 to 0.27 over 1.2×10^9 steps reached 71 in one of three seeds (the others ended at 77 and 76). A single hot restart from 71 (0.65 to 0.28) reached 69 within seconds. Cooling broke a floor that longer fixed-temperature runs did not, but these are few runs.

Annealing did not lower the best counts for $N \geq 24$. Cooling runs from $n = 8$: 113, $n = 9$: 139, $n = 10$: 151 and $n = 11$: 181, with several schedules and up to 1.5×10^9 steps per run, all ended where they started. So did a single hot run at $n = 12$: 191. At these sizes most proposals raise the count by 6 or more. A temperature low enough to keep the walk near the floor rejects almost all of them, and one high enough to move it destroys the structure.

VI. RESULTS

Records. Table I lists the twelve certified circuits and how each was obtained. The reductions relative to the published values grow with n: 4–10% for $n \leq 10$, 16–30% for $n = 11$–16, 51% for $n = 32$ and 66% for $n = 64$. The moduli are listed in Appendix A. All the starts are deterministic and take well under a second to build. The searches are seeded, and their outputs are stored.

Optimality for $n = 2$ and for $\mathbb{F}_2[y]/(y^2)$. For GF(4) ($N = 6$), the code $\mathcal{C}$ has dimension 22. Enumerating all 2^{22} coset elements gives minimum weight 17, attained by 1008 decompositions, with the next weights 18 (2576) and 21 (24 816). Hence $\mathcal{T}(2) = 17$, which the published circuits already attain [5, 6]. A second implementation, written independently, reproduced these numbers. The same enumeration shows that $\mathcal{T}(\mathbb{F}_2[y]/(y^2)) = 15$ (112 minimizers) and $\mathcal{T}(\mathbb{F}_2 \times \mathbb{F}_2) = 13$. The auditor agent reproduced both with its own code.

Small fields. Our certified values for $n \leq 5$ and $n = 9$ do not beat the published ones. They are $\mathcal{T}(3) \leq 29$, $\mathcal{T}(4) \leq 39$ (equal to Ref. [6]), $\mathcal{T}(5) \leq 58$ (published 49 [8]) and $\mathcal{T}(9) \leq 139$ (equal to Ref. [8]). Two of the small published values deserve comment.

For $n = 3$, Ref. [5] reports 23 and Ref. [8] 27. We could not reproduce either value for GF(8). More than 3000 FastTODD-type and greedy descents, from every field start we tried, end at exactly 29. These include an exact rebuild of the 40-column Karatsuba start that Ref. [8] reports. A Stern information-set-decoding search of the full coset found valid decompositions at weights 33 down to 29 at the rates a random-code model predicts. It found none of weight ≤ 28, where the model predicts about four ($P \approx 0.02$). Exact searches over decompositions invariant under several symmetry groups found nothing of weight ≤ 27. The public benchmark repository of Ref. [5] contains GF(2^n) circuits for $n \geq 4$ only, so the $n = 3$ circuit cannot be checked from source. The same Karatsuba start reduces to exactly 23 for some reducible moduli, for example $x^3 + 1$, where the algebra is $\mathbb{F}_2 \times$ GF(4) rather than a field. This suggests, but does not prove, that 23 is not a GF(8) value. We therefore keep 23 as the record in our verifier, and we do not use it as a base.

For $n = 5$, the published 49 [8] comes from a field start (the benchmark circuit uses $x^5 + x^2 + 1$), and we have no reason to doubt it. Our searches reach 58.

Bounds implied by published values. Theorem 1 also applies to published small values that we could not certify. $\mathcal{T}(5) = 49$ [8] would give $\mathcal{T}(10) \leq 3 \times 49 = 147$, below our 151. $\mathcal{T}(3) = 23$, if it is a GF(8) value, would give $\mathcal{T}(9) \leq 115$, $\mathcal{T}(12) \leq 161$ and $\mathcal{T}(15) \leq 207$. These bounds are conditional on those values. Our $n = 6$ record of 69 equals the bound 3×23 but does not depend on it.

VII. VERIFICATION

Frozen verifier. A candidate lists n, an irreducible modulus and the vectors u_r. The verifier checks irreducibility with Rabin's test, and then the criterion of Lemma 1 exactly, through the parities of all singles, pairs and triples. For each pair (i, j) it accumulates the XOR of the vectors containing both, so the cost is quadratic in the vector weight. By Lemma 1, a pass is a proof of correctness. A candidate wins only if it has fewer vectors than the record. Before the search started, the verifier's sources, the record table and the compiled checker were pinned by hash. They were re-pinned three times: after the record table was updated, after robustness fixes (Sec. VIII), and for the quadratic-cost tensor check. The verifier accepts all nine published AlphaTensor-Quantum decompositions for $n = 2$–10, accepts schoolbook CCZ constructions, rejects malformed input and exercises every verdict path. Its 37 tests include random property tests. All single-bit corruptions are rejected, and all zero-sum moves are accepted.

Independent circuit check. A separately written C program evaluates $4f_n(x) - \sum_r \ell_{u_r}(x) \bmod 8$ for all 2^{3n} inputs. It checks that the remainder is a diagonal Clifford phase, using a Walsh–Hadamard transform and a Möbius transform over $\mathbb{Z}_4$. The verifier runs it for $n \le 8$, and the two checks have never disagreed. The auditor agent (Sec. VIII) simulated every input for the records with $n = 6$, 7, 10, 11 and 12, the last over $2^{36} \approx 6.9 \times 10^{10}$ inputs, with no failure. So every record with $n \le 12$ is certified by full simulation as well.

Independent tensor checks. The coordinating session re-checked every new best with a third implementation. It uses numpy and shares no code with the verifier, and it builds $\sum_r u_r^{\otimes 3}$ and the target from scratch. Every record has zero mismatches among the N^3 entries, for instance 7 077 888 entries at $n = 64$. The auditor re-checked each record with its own tensor code and a randomized circuit test. Negative controls fail every check. These include single flipped bits and the seven nonzero vectors of a 3-dimensional subspace, which the weight-≤ 2 part of the check cannot see.

Novelty. We found no published T-count at or below ours for the twelve field sizes. The interpolation, CRT and curve constructions are classical in algebraic complexity [14–18, 20, 21]. We claim their combination with T-optimized blocks and the resulting circuits, not the constructions themselves. CRT-based multipliers have been used to lower Toffoli counts for binary fields with $n = 233$–571 [22]. We found no prior use of curve-based multiplication for T or Toffoli counts.

VIII. HOW THE RESULTS WERE FOUND

The work used two runs of a multi-agent system of language-model agents (Claude Sonnet 5.5, Anthropic), run as non-interactive coding sessions on one laptop with ten cores. A supervising session (Claude Opus 5.5) coordinated them. The agents shared a message board and could not modify the verifier, the record table or the board software. Every submission was frozen and verified, and the verifier, not the submitting agent, posted the verdict.

First run (subfields). Five agents took the roles of TODD-type reduction, bilinear construction, local search, large fields and audit. Six minutes after launch, the auditor found Ref. [8], which had lowered several of the records we had started from. The table was updated and re-frozen. Two wins certified in the minutes between the alert and the update ($n = 7$: 103, $n = 10$: 193) were revoked as superseded. The auditor also found three robustness bugs in the verifier and gaps in the board's identity model. None could produce a false win, and all were fixed. The large-field agent proposed Karatsuba over subfields 29 minutes after launch. The coordinating session suggested the interpolation generalization, and the first six composite records followed within 27 minutes.

Second run (places, curves, annealing). Six agents ran for four hours in 45-minute shifts. Their roles were reduction, three small-field agents ($n = 3$, 4 and 5, and their multiples), composition and audit. Two minutes after launch, the reduction agent lowered $n = 8$ to 113, and a minute later $n = 32$ to 571. A build for 192 variables reduced the first $n = 64$ circuit after 19 minutes. At seven minutes, the agent for $n = 4$ proposed CRT over places with local rings, giving the first prime-n records ($n = 11$: 189, $n = 13$: 228). At 18 minutes the dual-number places over GF(4) gave $n = 10$: 151. The composition agent wrote the dynamic program over places, and adding the p^2 place brought $n = 11$ and 13 to 181 and 219 after 29 minutes. The annealing chain for $n = 7$ ran after 39–44 minutes, and the genus-2 curve for $n = 64$ came after 50–53 minutes. Slow cooling for $n = 6$ came after 73 minutes (71) and a hot restart after 102 minutes (69). The agents also shared failures. The reheating chain did not transfer to $N \ge 24$, cooling stalled at $n = 8$–12, and fixed temperatures failed at $n = 6$. These reports shaped the directives of the coordinating session.

The auditor's role changed the paper as well as the record table. It checked the first version of this manuscript against the board and found stale claims and two errors. It also prompted the investigation of the $n = 3$ value, reading the benchmark repository and the original papers.

IX. OUTLOOK

Four directions follow. First, every improvement of a block in Table II lowers all the circuits that use it. For example, $\mathcal{T}(\mathbb{F}_2[y]/(y^3)) \le 24$ would lower $n = 7$, 11 and 13, and $\mathcal{T}(\mathrm{GF}(16)) \le 38$ would lower all eight circuits that contain GF(16) blocks. Second, the curve method applies whenever the line runs out of points. A

certified $n = 44$–60 series would test the cost model. Third, annealing broke the floors that reduction left at $n = 6$ and 7 but none for $N \geq 24$; better move sets are needed there. Fourth, lower bounds remain far away. For Hadamard-free phase circuits with clean ancillas, and hence for multipliers of our form $H_c D H_c$, Ref. [23] proves $\mathcal{T}(n) \geq 6n + 1$ and recovers $\mathcal{T}(2) = 17$. Our upper bounds are 17.8 T gates per bit at $n = 32$ and 19.6 at $n = 64$. Exhaustive enumeration is infeasible already at $n = 3$, where $\dim \mathcal{C} = 382$. The status of $\mathcal{T}(3)$ (23, 27 or 29) is open.

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We thank the authors of Refs. [5–8] for publishing benchmark values and, in the case of Ref. [6], their decompositions (CC-BY 4.0), which serve as base cases. The circuits were found by an autonomous multi-agent system built on Claude Sonnet 5.5 (Anthropic). The verification software, the analysis, and this manuscript were produced by Claude Opus 5.5 (Anthropic).

DATA AVAILABILITY

All twelve record decompositions, the verifier, its tests, the independent checkers, the construction scripts and the reduction and annealing programs are in the project repository. The composed starts are deterministic. The reductions and annealing runs are seeded, and their outputs are stored with the commands that produced them.

Appendix A: Construction data

Each record is specified by its modulus P_n and the construction in Table I. The decompositions, lists of vectors in $\mathbb{F}_2^{3n}$, are in the repository.

TABLE III. Moduli of the certified circuits.

n	P_n	n	P_n
6	$x^6 + x + 1$	13	$x^{13} + x^5 + x^2 + x + 1$
7	$x^7 + x + 1$	14	$x^{14} + x^5 + 1$
8	$x^8 + x^4 + x^3 + x + 1$	15	$x^{15} + x + 1$
10	$x^{10} + x^3 + 1$	16	$x^{16} + x^5 + x^3 + x + 1$
11	$x^{11} + x^2 + 1$	32	$x^{32} + x^7 + x^3 + x^2 + 1$
12	$x^{12} + x^3 + 1$	64	$x^{64} + x^{11} + x^2 + x + 1$

[1] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A **71**, 022316 (2005).
[2] D. Litinski, Magic state distillation: Not as costly as you think, Quantum **3**, 205 (2019).
[3] M. Amy and M. Mosca, T-count optimization and Reed–Muller codes, IEEE Trans. Inf. Theory **65**, 4771 (2019); arXiv:1601.07363.
[4] L. E. Heyfron and E. T. Campbell, An efficient quantum compiler that reduces T count, Quantum Sci. Technol. **4**, 015004 (2019); arXiv:1712.01557.
[5] V. Vandaele, Lower T-count with faster algorithms, Quantum **9**, 1860 (2025); arXiv:2407.08695.
[6] F. J. R. Ruiz *et al.*, Quantum circuit optimization with AlphaTensor, Nat. Mach. Intell. (2025), doi:10.1038/s42256-025-01001-1; arXiv:2402.14396; decompositions at https://github.com/google-deepmind/alphatensor_quantum.
[7] K. Khoruzhii, P. Gelß, and S. Pokutta, Tensor decomposition for non-Clifford gate minimization, arXiv:2602.15285 (2026).
[8] D. Fisher, V. Khrulkov, M. Saygin, I. Oseledets, and S. Straupe, LLM-guided evolutionary search for algebraic T-count optimization, arXiv:2603.29894 (2026).
[9] M. Amy, D. Maslov, and M. Mosca, Polynomial-time T-depth optimization of Clifford+T circuits via matroid partitioning, IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst. **33**, 1476 (2014).
[10] G. Banegas, D. J. Bernstein, I. van Hoof, and T. Lange, Concrete quantum cryptanalysis of binary elliptic curves, IACR Trans. Cryptogr. Hardw. Embed. Syst. **2021**(1), 451 (2021).
[11] I. van Hoof, Space-efficient quantum multiplication of polynomials for binary finite fields with sub-quadratic Toffoli gate count, Quantum Inf. Comput. **20**, 721 (2020).
[12] N. Yosri, D. Gavinsky, and D. Maslov, Asymptotic yet practical optimization of quantum circuits implementing GF(2^m) multiplication and division operations, arXiv:2511.20618 (2025).
[13] V. Vandaele, Quantum binary field multiplication with subquadratic Toffoli gate count and low space–time cost, arXiv:2501.16136 (2025).
[14] A. Karatsuba and Yu. Ofman, Multiplication of many-digital numbers by automatic computers, Dokl. Akad. Nauk SSSR **145**, 293 (1962).
[15] A. L. Toom, The complexity of a scheme of functional elements realizing the multiplication of integers, Soviet Math. Dokl. **3**, 714 (1963).
[16] S. Winograd, On multiplication in algebraic extension fields, Theor. Comput. Sci. **8**, 359 (1979).
[17] P. Bürgisser, M. Clausen, and M. A. Shokrollahi, *Algebraic Complexity Theory* (Springer, Berlin, 1997).
[18] D. V. Chudnovsky and G. V. Chudnovsky, Algebraic complexities and algebraic curves over finite fields, J. Complexity **4**, 285 (1988).

[19] I. E. Shparlinski, M. A. Tsfasman, and S. G. Vladut, Curves with many points and multiplication in finite fields, in *Coding Theory and Algebraic Geometry*, Lecture Notes in Mathematics Vol. 1518 (Springer, Berlin, 1992), pp. 145–169.

[20] H. Randriambololona, Bilinear complexity of algebras and the Chudnovsky–Chudnovsky interpolation method, J. Complexity **28**, 489 (2012); arXiv:1107.0336.

[21] S. Ballet, J. Chaumine, J. Pieltant, M. Rambaud, H. Randriambololona, and R. Rolland, On the tensor rank of multiplication in finite extensions of finite fields and related issues in algebraic geometry, Russian Math. Surveys **76**, 29 (2021); arXiv:1906.07456.

[22] S. Kim, I. Kim, S. Kim, and S. Hong, Toffoli gate count optimized space-efficient quantum circuit for binary field multiplication, Quantum Inf. Process. **23**, 330 (2024).

[23] A. R. Mazumder, Exact T-counts of CCZ layers from an isotropy bound, arXiv:2610.01024 (2026).